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## How Goals-Based Portfolio Theory Came to Be

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The following is excerpted from *Goals-Based Portfolio Theory* (<https://www.wiley.com/en-us/Goals+Based+Portfolio+Theory-p-9781119906100>) by Franklin J. Parker, CFA, published this year by Wiley (<https://www.wiley.com/en-us>).

"I've heard people compare knowledge of a topic to a tree. If you don't fully get it, it's like a tree in your head with no trunk – when you learn something new about the topic there's nothing for it to hang onto, so it just falls away." –Tim Urban

When presented a choice between multiple possibilities, which one should you choose? This simple question has perplexed many a human being. Modern economics found its beginning with an attempt to answer this basic question. The wealthy class of Europe had quite a bit of time on their hands, and, as it turned out, they enjoyed gambling on games of chance. The Renaissance had shifted the traditional view of these games – rather than simply accept randomness, some of these aristocrats began to analyze the games mathematically in an attempt to understand their randomness. It was not through any pure mathematical interest, of course, but rather an attempt to gain an edge over their fellow gamblers and thereby collect more winnings!

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The thinking of the time coalesced around a central idea: expected value theory. Expected value theory stated that a gambler should expect to collect winnings according to the summed product of the gains or losses and the probabilities of those outcomes (i.e.,  $\sum_i p_i v_i$ , where  $p$  is the probability of gaining/losing  $v$ , and  $i$  is the index of possible outcomes). If, for example, you win \$1 every time a six-sided die rolls an even number, and you lose \$1 when it rolls odd, then the expected value of the game is  $1/2 \times \$1 + 1/2 \times (-\$1) = \$0$ .

In 1738, Daniel Bernoulli challenged that idea

(<https://blogs.cfainstitute.org/investor/2021/02/17/bernoullis-prisoners-dilemma-a-goals-based-perspective/>). As a thought experiment he proposed a game: a player is given an initial pot of \$2, and a coin is flipped repeatedly. For every heads, the player doubles their money and the game continues until the coin lands on tails. When tails comes up, the player collects winnings of  $\$2^n$ , where  $n$  is the number of times the coin was flipped, and the game is over. Bernoulli's question is, how much should you pay to play this game?

Expected value theory fails us here because the payoff of the game is infinite! Clearly no one would pay an infinite amount of money to play the game, but why? Bernoulli's answer is our first glimpse of a marginal theory of utility – a theory that would come to support all modern economics:

"Thus it becomes evident that no valid measurement of the value of a risk can be obtained without consideration being given to its utility, that is to say, the utility of whatever gain accrues to the individual or, conversely, how much profit is required to yield a given utility. However it hardly seems plausible to make any precise generalizations since the utility of an item may change with circumstances. Thus, though a poor man generally obtains more utility than does a rich man from an equal gain, it is nevertheless conceivable, for example, that a rich prisoner who

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possesses two thousand ducats but needs two thousand ducats more to repurchase his freedom, will place a higher value on a gain of two thousand ducats than does another man who has less money than he.”



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The idea that humans do not value changes in wealth linearly, but rather find less value in the next ducat than they found in the first, launched the entirety of modern economics. Bernoulli went on to propose a logarithmic function for the utility of wealth — diminishing as the payoff grows. This, of course, solved the paradox. People are not willing to pay an infinite amount to play the game because they do not have infinite utility for that wealth. The value of each subsequent dollar is less than the previous one — that is the essence of marginal utility, and the foundation of modern economics.

Of more interest to this discussion, however, is that Bernoulli also gives a first glimpse of a goals-based theory of utility! Bernoulli points out that we must think of what it is the wealth *can do for us*, rather than the absolute value of that wealth. In other words, it is not the cash that we care about, but rather what that cash represents in the real world: freedom from prison in Bernoulli's Prisoner's case, and transportation, housing, leisure, food, and so on, for the rest of us. What you wish to do with the money is an important consideration to how much you would pay to play Bernoulli's game. This idea is echoed by Robert Shiller, winner of the 2013 Nobel Prize in Economics: "Finance is not merely about making money. It is about achieving our deep goals and protecting the fruits of our labor." In short, investing is never done in the abstract! Investing is — and always has been — goals-based.

It would be another two centuries before the theory underpinning rational choices was developed. John von Neumann and Oskar Morgenstern authored *The Theory of Games and Economic Behavior*

(<https://press.princeton.edu/books/paperback/9780691130613/theory-of-games-and-economic-behavior>) in 1944, which has become the foundation upon which all theories of rational choice are built. Von Neumann was a mathematician (and a brilliant one at that), so their additional contribution — beyond the actual foundational ideas — was to apply a mathematical rigor to the theory of human choice.



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In 1948, Milton Friedman (later to win the 1976 Nobel prize in economics) and L. Savage explored the implications of von Neumann and Morgenstern's rational choice theory to an economic conundrum: why do people buy both insurance and lottery tickets? Rational choice theory would generally expect individuals to be variance-averse, so the fact that people express preferences for both variance-aversion and variance-affinity in the same instance is troubling. This has since become known as the Friedman-Savage paradox, and their solution was that the utility curve of individuals must not contain one curve, but many interlinked curves. That is, it must be "squiggly," shifting between concave and convex across the wealth/income spectrum — known as the double-inflection solution. (When a utility curve is convex, individuals are variance-averse, and when concave, individuals are variance-affine. Friedman and Savage's solution is clever and was, in fact, reiterated by Harry Markowitz's 1952 paper "[The Utility of Wealth](https://www.jstor.org/stable/1825964)." (<https://www.jstor.org/stable/1825964>)) As it turns out, this is also a proto-goals-based solution, as the goals-based utility curve is also "squiggly," moving from concave to convex across the spectrum of wealth.

Even more than the method it contained, Markowitz's other monumental 1952 paper "[Portfolio Selection](https://onlinelibrary.wiley.com/doi/10.1111/j.1540-6261.1952.tb01525.x)" (<https://onlinelibrary.wiley.com/doi/10.1111/j.1540-6261.1952.tb01525.x>) was the first serious application of statistical techniques to investment management. Prior to Markowitz, investment management was a bottom-up affair: a portfolio was merely the aggregate result of many individual decisions about securities. Benjamin Graham's *The Intelligent Investor* (<https://www.harpercollins.com/products/the-intelligent-investor-rev-ed-benjamin-graham?variant=32123199782946>) is a characteristic example (though by no means the only approach at the time). Nowhere in his classic text is Graham concerned with how the various investments within a portfolio interact to create the whole. Rather, it is the job of the investor to simply identify attractive opportunities and add them to their portfolio, replacing ideas that have been played out. The portfolio, then, is the aggregate result of these many unrelated decisions.

By applying statistical techniques to the portfolio and suggesting investors evaluate individual investment opportunities within the context of the portfolio as a whole, Markowitz showed that (a) investors could get more done with the same amount of money, and (b) quantitative methods could have a significant role to play in investment management. Both of those breakthroughs hold to this day.



(<https://www.tandfonline.com/toc/ufaj20/current>)

Markowitz was not the only voice in the debate, of course. In the same year Markowitz published his breakthrough paper, Roy published "[Safety First and the Holding of Assets](https://www.jstor.org/stable/1907413)." (<https://www.jstor.org/stable/1907413>) Ironically, Roy's paper looks much more like what we have come to know as modern portfolio theory. Indeed, nowhere in Markowitz's original paper does the now-familiar efficient frontier appear, but Roy's has not only a proto-efficient frontier, but the capital market line, and an early version of the Sharpe ratio to boot! What's more, Roy's entire analysis is dedicated to the idea that individuals never have a "sense of security" in the real world. That is, never do people have all the information, nor are they always seeking to simply maximize profits. Rather, individuals are attempting to maximize profits *and* avoid the landmines that could well destroy their hard-won progress:

“A valid objection to much economic theory is that it is set against a background of ease and safety. To dispel this artificial sense of security, theory should take account of the often close resemblance between economic life and navigation in poorly charted waters or maneuvers in a hostile jungle. Decisions taken in practice are less concerned with whether a little more of this or of that will yield the largest net increase in satisfaction than with avoiding known rocks of uncertain position or with deploying forces so that, if there is an ambush round the next corner, total disaster is avoided. If economic survival is always taken for granted, the rules of behavior applicable in an uncertain and ruthless world cannot be discovered.”



([https://www.cfainstitute.org/en/research/survey-reports/future-of-work-content/?s\\_cid=dsp\\_FOSLConsult\\_EnterprisingInvestor](https://www.cfainstitute.org/en/research/survey-reports/future-of-work-content/?s_cid=dsp_FOSLConsult_EnterprisingInvestor))

Markowitz's line of thinking also held considerable appeal to the well-funded pension schemes and insurance companies of the 1950s, 1960s, and 1970s. These institutions had the financial ability and interest to fund research that spoke to how they might better achieve the objectives of their pensioners and shareholders. Hence, portfolio theory developed with institutions — not individuals — in mind. For many years, it was assumed that the differences were so negligible as to be not worth exploring. After all, statistics is statistics whether the portfolio is worth \$1 billion or \$100,000.

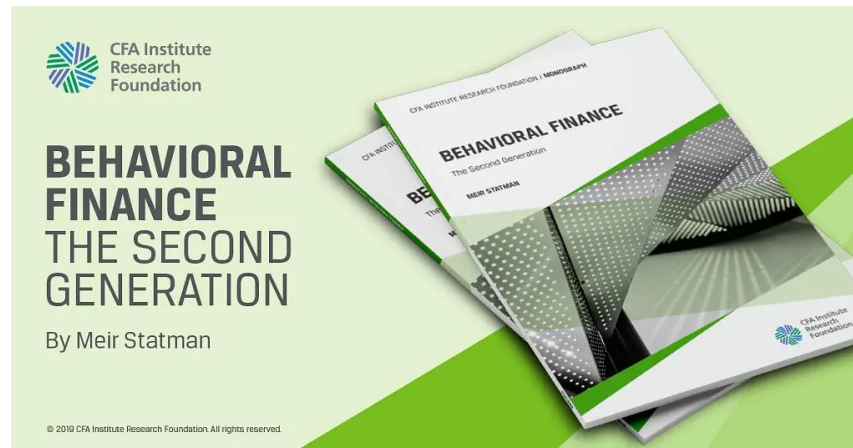
Yet, as we now understand, there are *substantial* differences between a \$1 billion pension fund and a \$100,000 investment account. Surprisingly, it wasn't until 1993 — three years after Markowitz collected his well-deserved Nobel prize — that Robert Jeffrey and Robert Arnott fired this first salvo at institutionally oriented portfolio theory. Their paper was titled “[Is Your Alpha Big Enough to Cover Its Taxes?](https://jpm.pm-research.com/content/19/3/15)” (<https://jpm.pm-research.com/content/19/3/15>) and it opens:

“Much capital and intellectual energy has been invested over the years in seeking to make portfolio management more efficient. But most of this effort has been directed at *tax-exempt* investors such as pension funds, foundations, and endowments, even though taxes are a major consideration for owners of approximately two-thirds of the marketable portfolio assets in the United States.” (Emphasis is in the original)

The authors go on to discuss how taxable investors can think about tax-drag as a central concern of their investment strategy, rather than as an afterthought. In the historical development of goals-based portfolio theory, their research was among the first to systematically redress a difference between individual investors and the investors for whom portfolio theory was developed, namely institutions. It was the first clue that, yes, portfolio results might legitimately be different for taxable investors, even if the statistical tools were the same.

Of course, by the early 1990s, the behavioral economics revolution was in full swing. A decade before, in 1979, [Daniel Kahneman and Amos Tversky presented the results of their psychological research](https://www.jstor.org/stable/1914185) (<https://www.jstor.org/stable/1914185>), which had considerable bearing on economics. In short, they found that people feel the pain of financial loss more strongly than they feel the pleasure of financial gain, and when

coupled with their further observation that people seem not to weight probabilities objectively, we have their full theory, known as cumulative prospect theory (CPT), for which Kahneman would later win the 2002 Nobel Prize in Economics.



([https://www.cfainstitute.org/en/research/foundation/2019/behavioral-finance-the-second-generation/?s\\_cid=dsp\\_eilnHouseADS\\_CFA\\_EI\\_banner\\_1x1](https://www.cfainstitute.org/en/research/foundation/2019/behavioral-finance-the-second-generation/?s_cid=dsp_eilnHouseADS_CFA_EI_banner_1x1))

Expanding their work, Richard Thaler (winner of the 2017 Nobel Prize in Economics) developed the concept of mental accounting. He proposed that people mentally subdivide their wealth into different “buckets,” and each bucket carries a different risk tolerance. Mental accounting also resolved some behavioral conundrums, like the Friedman-Savage paradox. If people have some of their wealth mentally dedicated to survival objectives and some of their wealth dedicated to aspirational objectives, then these differing risk tolerances will yield people who buy both insurance and lottery tickets. Rather than one interlocking “squiggly” utility curve, mental accounting suggests that people have many separate utility curves.

Mental accounting was also a throwback to the ideas of psychologist Abraham Maslow (<https://psycnet.apa.org/record/1943-03751-001>). People have multiple psychological and physical needs at any given moment: food, shelter, safety, a sense of belonging, self-esteem, and so on. While humans may have their physical needs met, they will still seek to fulfill more abstract psychological needs. Maslow proposed that these needs are fulfilled in a sort of hierarchy, with physiological needs being fulfilled first (food, water, shelter), and psychological needs fulfilled only after those physiological needs are met. This concept is usually presented as a pyramid, although Maslow was himself not so rigid, proposing that individuals will tend to prioritize these needs differently across the course of their life. Toward the end of our lives, for example, Maslow suggests we have a strong need for esteem and self-actualization, with more physiological needs a priority in earlier life. Although, if something happens that destroys an individual’s sense of physiological safety, the higher objectives will collapse as the individual attempts to fulfill her base needs.

Mental accounting was foundational to goals-based investing because it was the first acknowledgment and theoretical treatment of investors who divvy their wealth across multiple objectives, reflective of Maslow’s observation. Yet in Thaler’s early treatment, mental accounting was considered a cognitive bias and therefore irrational. It violated the basic premise that money is fungible — you can swap a dollar here for a dollar there — and as Markowitz showed, investors are best served by considering a portfolio of investments from the top down. Mental accounting, by contrast, was seen as a return to a bottom-up approach. So, though people may *behave* in a way that treats money differently depending on which mental account it is in, people *shouldn’t* do that from the perspective of traditional economic theory. It was almost another two decades before Jean L. P. Brunel took up the question and demonstrated that this subdivision of wealth across multiple accounts

([https://www.researchgate.net/publication/240316087\\_How\\_SubOptimal-If\\_at\\_All-Is\\_Goal-Based\\_Asset\\_Allocation](https://www.researchgate.net/publication/240316087_How_SubOptimal-If_at_All-Is_Goal-Based_Asset_Allocation)) — mental or actual — is not necessarily irrational or suboptimal. Thanks to Brunel’s work, there are now two uses of the term *mental*

*accounting*. The first is the cognitive bias wherein people do not treat money as

fungible. The second is the observation that people tend to dedicate their wealth toward different goals, and, in response to those differing objectives, they tend to pursue differing types of investments and strategies. While the former is irrational, the latter is not. Goals-based theory is concerned with the latter, as it expects money to be fungible.



([https://www.cfainstitute.org/en/research/foundation/2022/rflr-stock-buybacks/?s\\_cid=dsp\\_eilnHouseADS\\_CFA\\_EI\\_banner\\_1x1](https://www.cfainstitute.org/en/research/foundation/2022/rflr-stock-buybacks/?s_cid=dsp_eilnHouseADS_CFA_EI_banner_1x1))

The final idea that helped to coalesce the goals-based framework came in 2000 from Hersh Shefrin and Meir Statman (<https://www.jstor.org/stable/2676187>), who developed behavioral portfolio theory (BPT). BPT resurrects Roy's safety-first criterion and, in contrast to modern portfolio theory's risk-is-variance paradigm, BPT suggests that risk is the probability of failing to achieve some minimum required return. Said another way, BPT suggests that risk is the probability that you do not achieve your goal. When I think about my own life goals, this is *exactly* how I would define risk! In BPT, an investor builds a portfolio to balance expected return and the probability of failure, which is an analog to the mean-variance efficient frontier.

Despite its insight, BPT never gained mainstream acceptance. In 2010, however, Meir Statman teamed up with Sanjiv Das, Jonathan Scheid, and Harry Markowitz (<https://www.jstor.org/stable/27801487>) to merge the insights of behavioral portfolio theory with the framework of modern portfolio theory ([https://www.researchgate.net/publication/275907078\\_Portfolios\\_for\\_Investors\\_Who\\_Want\\_to\\_Reach\\_Their\\_C\\_Variance\\_Efficient\\_Frontier](https://www.researchgate.net/publication/275907078_Portfolios_for_Investors_Who_Want_to_Reach_Their_C_Variance_Efficient_Frontier)). They showed that the probability of failing to reach some threshold return is mathematically synonymous with mean-variance optimization, so long as short-selling and leverage were unconstrained (which is a common mean-variance assumption). In that context, an investor can simply declare the maximum probability of failure they are willing to accept for a given account, that metric can be "translated" into a risk-aversion parameter, and portfolio optimization can proceed in the traditional mean-variance way. Additionally, these authors showed, with considerable rigor, that the subdivision of wealth into multiple accounts is not necessarily irrational nor inefficient (an echo of Brunel's 2006 result).

My own entrée into the ideas of goals-based investing came in 2014 when, in the vertiginous years after 2008, I was left wondering whether the traditional methods of portfolio management were still relevant. Experience taught me — like it taught so many in 2008 — that the math is simply different for individuals who have specific objectives to achieve within a specified period of time. I felt quite silly for waving off previous client protestations of portfolio losses. They intuitively understood what I explained away with flawed theory. Insurance companies can wait five years for their risk to be rewarded, but individuals who plan to retire simply cannot, and those who are living off of portfolio withdrawals can even less afford to wait. After that experience, I had one central question: *How much can you lose in an investment portfolio before you've lost too much?* Markets, of course, come back — that was never my concern. My concern was whether they come back *in time* for my clients to achieve their goals. Again, I discovered what others had before me: portfolio theory for individuals is legitimately

different than portfolio theory for institutions. After realizing that no one had an answer to my basic question, I developed my own answer, resulting in my first peer-reviewed publication (<https://jwm.pm-research.com/content/17/3/68>).



([https://www.cfainstitute.org/en/research/industry-research/t-shaped-teams/?s\\_cid=dsp\\_eiInHouseADS\\_CFA\\_EI\\_banner\\_1x1](https://www.cfainstitute.org/en/research/industry-research/t-shaped-teams/?s_cid=dsp_eiInHouseADS_CFA_EI_banner_1x1))

My basic question post-2008 is illustrative of another aspect of goals-based portfolio theory. While it is about optimizing portfolios in a way that maximizes the probability of goal achievement, the whole ethos is about more than that. At its core, goals-based portfolio theory is about organizing your resources to maximize the probability of achieving your goals *given real-world constraints*. It is the “real-world constraints” component that has been so often neglected by traditional portfolio theory. It would be nice if investors had access to unlimited leverage and short-selling, but they do not! It would be very nice if investment returns were Gaussian, but they are not. Pretending as though absurd assumptions are reality, then acting surprised when practice mismatches theory, is just plain silliness. While we must accept that theory is not reality, we can do better than a theory that could never be reality. More than anything, investors need a theory that is *useful*.

Recognizing this, Brunel coalesced these various ideas into a whole in his book *Goals-Based Wealth Management* (<https://www.wiley.com/en-us/Goals+Based+Wealth+Management:+An+Integrated+and+Practical+Approach+to+Changing+the+Structure+p-9781118995907>), which addresses how practitioners might tackle the problems of organizing resources for investors with goals to achieve. Having spent many decades at the beating heart of the financial system, serving real people with real goals to achieve, Brunel’s work is uniquely positioned at the intersection of the “big world” and the client’s world. How firms can systematize these ideas into scalable solutions is no small question, and his book addresses these practical challenges, as well.

Once the goals-based definition of risk gained wider acceptance, the next major question was how investors should allocate across their various mental accounts. The assumption for many years was that this allocation across goals was already done by the investor, so the practitioner’s job was to organize the investments within each goal in the optimal way. However, to expect investors to rationally allocate wealth across goals is somewhat naïve. To be fair, there are currently several approaches in the literature. In my book (<https://www.wiley.com/en-us/Goals+Based+Portfolio+Theory-p-9781119906100>), I present my solution and briefly address my critiques of some of the other major approaches, but I do not want to sound as though this is a settled question. Other researchers may yet present a better solution than mine, and in that case, I will yield the ground I claim here. Though solved to my mind, how investors should allocate across goals is still an open question.

**For more from Franklin J. Parker, CFA, check out *Goals-Based Portfolio Theory* (<https://www.wiley.com/en-us/Goals+Based+Portfolio+Theory-p-9781119906100>) and follow him at *Directional Advisors* (<https://directionaladvisors.com>).** By continuing to use the site, you agree to the use of cookies. [more information](#)

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Thank you for such a wonderful and insightful posting. This concisely summarizes history of developments of GBI theories. I have researched and read a few papers and books about those theories, and now I feel like things are putting together so much more than before. Let me head into my kindle to further read your book. It should help me in developing and bringing GBI theories into the wealth tech or wealth management industry.

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